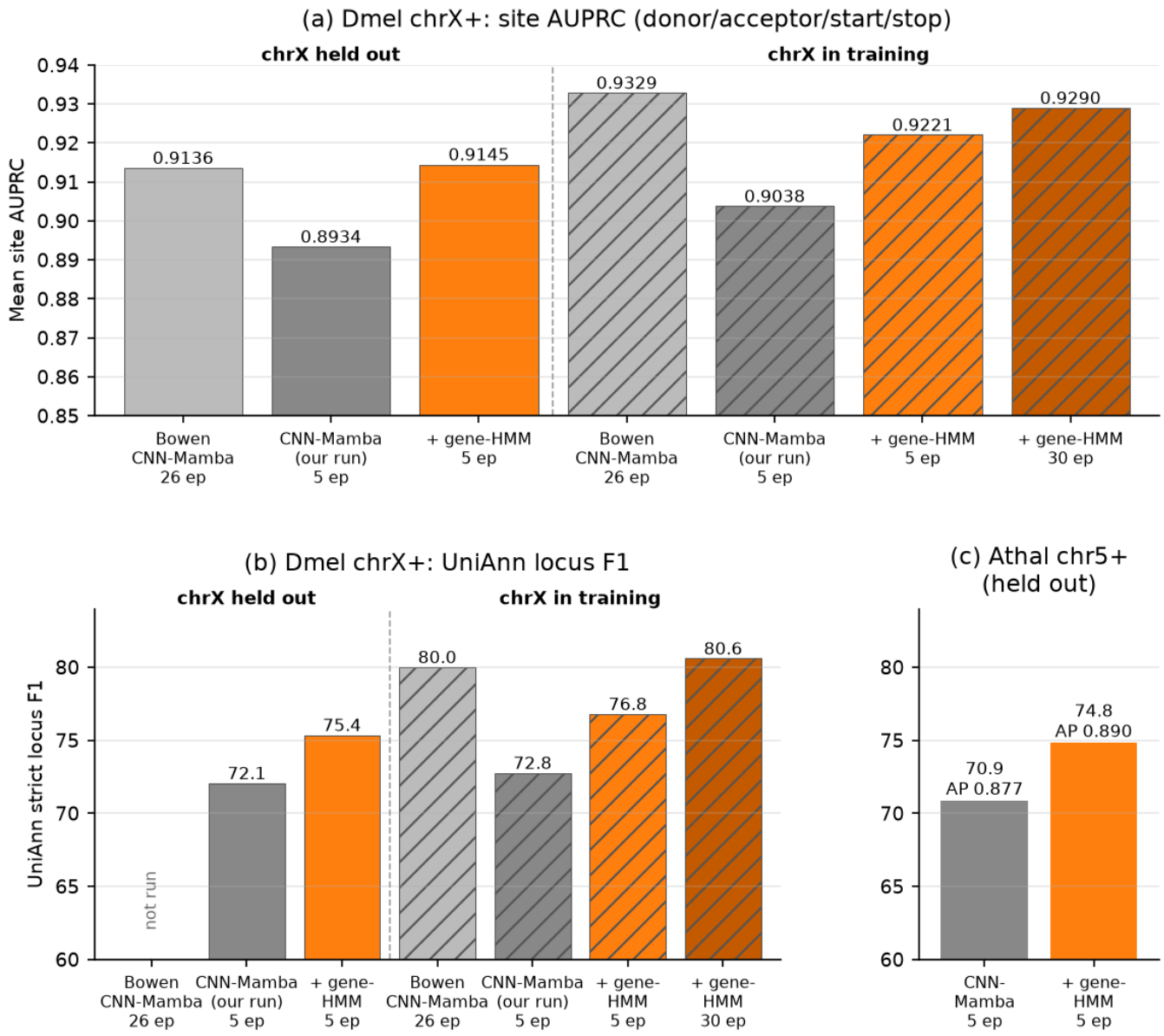
Gene-HMM (CRF) layer for CNN-Mamba: handoff notes

Status as of 2026-10-21. Author: Hibiki (with Claude Code). Code: branch `gene-hmm-isoforms` of https://github.com/hibiki-kato/cnn-mamba (a fork of `bowencccc/cnn-mamba`, branched from `main` at 809f743).

TL;DR

- We add a small, differentiable **gene-structure CRF** on top of the existing CNN-Mamba heads and train it **end-to-end** with the heads. At inference, the CRF's **posterior marginals**, not the raw heads, become the site and per-base frame scores fed to UniAnn.
- In every setting we ran, this improves UniAnn **strict locus F1** over the same backbone without the HMM:
  - Dmel chrX+: +2.2 to +4.1.
  - Athal chr5+: +4.0 (70.9 → 74.8).
- These gains are against **our own 5-epoch baseline**, which is under-trained. Against your fully trained model the gain is much smaller:
  - Held-out site AUPRC is a tie (0.9145 vs 0.9136).
  - Your 80.0 locus F1 is in the chrX-in-training regime. In that regime at 5 epochs we get 72.8 without the HMM and 76.8 with it; trained as long as yours (30-ep cosine), the gene-HMM gets **80.6** (section 0). With one seed, we read this as a tie or a small gain.
  - So a large part of the +3 to +4 at 5 epochs is faster convergence. The HMM's clearest effect is on locus precision.
- The gain is mostly **locus precision**: the HMM model predicts fewer false loci. Site AUPRC barely shows it; for Athal, AUPRC rises only by +0.013.
- **End-to-end training is required.** Fitting the same CRF post hoc on a frozen baseline makes results *worse* in both species.
- The raw heads of the HMM-trained model are not consistently better than the baseline. The gain comes from **decoding through the HMM posterior**.
- Caveats are in section 6:
  - The posterior is too sure in the low tail, so it vetoes 1–5% of true sites.
  - The `pr_all.pl` converter multiplies by P(coding) twice.
  - Cosine schedule plus early stopping is a bad combination (section 5).

0. Headline: your CNN-Mamba vs + gene-HMM



The gene-HMM rows use the best setting: scratch, HMM loss weight 1, decoded through the HMM posterior. The 30-epoch `chrxtrain` row is the one to compare with your 80.0. All rows use the same converter (`pr_all.pl`), the same `uniann-pr45`, and `gffcompare --strict-match -e 0` against the same plus-strand reference.

Model	Test chrom in training?	Epochs	Donor AP	Acc. AP	Start AP	Stop AP	**Mean AP**	**Locus F1** (Sn / Pr)
Dmel chrX+								
Your CNN-Mamba b010 (4-state β=0.10), `heldout`	no	26 (ES)	0.9570	0.9577	0.8547	0.8850	0.9136	not run (checkpoint not here)
Your CNN-Mamba b010, `chrxtrain` (annotation regime)	**yes**	26 (ES)	0.9684	0.9647	0.8908	0.9075	0.9329	80.0 (80.4 / 79.6)[1]
Same recipe, our run (arm B)	no	5	0.9478	0.9413	0.8191	0.8654	0.8934	72.1 (76.5 / 68.1)
+ gene-HMM (arm A, HMM posterior)	no	5	0.9604	0.9582	0.8502	0.8894	**0.9145**	**75.4** (75.9 / 74.8)
+ gene-HMM, `chrxtrain` (annotation regime, HMM posterior)	yes	5	0.9654	0.9625	0.8625	0.8982	0.9221	**76.8** (79.0 / 74.7)
same model, raw heads	yes	5	0.9546	0.9507	0.8594	0.8992	0.9160	73.9 (78.8 / 69.6)
Same recipe, our run (arm B), `chrxtrain`	yes	5	0.9446	0.9419	0.8389	0.8898	0.9038	72.8 (76.1 / 69.7)
+ gene-HMM, `chrxtrain`, 30-ep cosine (best val epoch 18, HMM posterior)	yes	30	0.9696	0.9684	0.8700	0.9080	0.9290	**80.6** (81.4 / 79.8)
same model, raw heads	yes	30	0.9639	0.9624	0.8844	0.9090	0.9299	76.9 (81.0 / 73.2)
Athal chr5+								
Same recipe (arm B)	no	5	0.9510	0.9364	0.8233	0.7957	0.8766	70.9 (73.1 / 68.8)
+ gene-HMM (arm A, HMM posterior)	no	5	0.9532	0.9468	0.8344	0.8254	**0.8899**	**74.8** (73.5 / 76.2)

[1] From `results/joint_frame_grid_v1/summary/`. Re-running your delivered scores through the corrected `pr_all.pl` gives 79.6.

How to read this table:

- **Your UniAnn number (80.0) comes from the chrX-in-training model.** This is the intended annotation setting, not leakage.
  - Training never sees FlyBase. The splice/start/stop, frame and path losses all use EviAnn pseudo-labels; FlyBase is used only for validation AP on chr3L and for evaluation.
  - Training on the target chromosome's EviAnn labels is exactly how the model would annotate a new genome.
- **Like-for-like in the annotation regime (chrX in training, 5 epochs each):** adding the gene-HMM gives 72.8 → 76.8 locus F1 (+4.0), again mostly from precision (69.7 → 74.7). This is the same size of gain as held-out Dmel (+3.3) and Athal (+4.0).
- **With matched training length the gene-HMM reaches 80.6 vs your 80.0** (30-ep cosine, no early stopping, best validation epoch 18).
  - The 5-epoch gap (76.8) was under-training on the training chromosomes themselves. Validation AP on chr3L did not move after epoch 5 (0.9235 vs 0.9234), but chrX site AP rose from 0.916 to 0.930 (raw heads) because chrX's EviAnn labels are in training.
  - Locus precision is the same as yours (79.8 vs 79.6); sensitivity is +1.0 (81.4 vs 80.4).
  - This is one seed. Your 80.0 is the best of five settings (74.1–80.0) picked on this test, so we read this as a tie or a small gain, not a clear win.
  - Decoding through the HMM posterior is still what matters: the same model's raw heads give 76.9.
  - After epoch 18 the model overfits: validation loss 0.013 → 0.025 and validation HMM AP 0.923 → 0.885 by epoch 30. The HMM posterior degrades faster than the raw heads.
- **Your held-out model has no UniAnn number.** It is not needed for the annotation comparison; site AUPRC is compared held-out.
- **Site AUPRC on held-out chrX** is a tie: 0.9145 (gene-HMM, 5 epochs) vs 0.9136 (your model, 26 epochs).
- **Our arm B is under-trained.** Our 5-epoch run of your recipe reaches only 0.8934 AP and 72.1 locus F1 (validation AP 0.895 vs 0.919 for your run). The +3.3 (Dmel) and +4.0 (Athal) locus-F1 gains at 5 epochs are therefore largely faster convergence; the gene-HMM model converges in about 5 epochs on validation AP.

1. What we are doing

Current pipeline:

1. CNN-Mamba predicts, independently at each position, the site probabilities (donor/acceptor/start/stop) and the per-base joint-frame probabilities (non-CDS, frame 0/1/2).
2. UniAnn's Viterbi assembles gene models from those scores.

Problem: the heads are trained position by position. Nothing in training tells them that a donor must be followed by an acceptor, that the frame must survive an intron, or that a stop ends the CDS. Gene-structure consistency is enforced only at decoding time.

Idea (as in Tiberius): put a gene-structure HMM inside the network, so that training is penalised whenever the *whole parse* is wrong, not just individual positions.

State space (single strand, model orientation)

State	Meaning
`IR`	intergenic / UTR / any non-coding base
`E0, E1, E2`	coding base at codon position 0/1/2
`I0, I1, I2`	intron; `I_k` re-enters the CDS at codon position k (carries the phase across the intron)

Allowed transitions are gated by sequence motifs. Each one is scored by the existing candidate head at that position:

Transition	Allowed when	Score
`IR → E0` (start)	ATG starts at t	start head at t
`E2 → IR` (stop)	stop codon at t-3..t-1	stop head at t-3
`E_k → I_{k+1}` (donor)	GT starts at t	donor head at t
`I_k → E_k` (acceptor)	AG at t-2..t-1	acceptor head at t-2
`E0→E1→E2→E0`, `IR→IR`, `I→I`	always (except `E2→E0` across an in-frame stop)	0

- Emissions come from a new 5-class **state head** (non-CDS / frame0 / frame1 / frame2 / intron). The three intron states share the intron class.
- The HMM itself has only **9 learnable scalars**: a weight and a bias for each of the four boundary types, plus one emission weight. See section 2 for the equations.
- The transition structure is fixed. The site heads supply the position-specific transition scores.
- Code: `src/cnn_mamba/gene_hmm.py`.
  - The partition function uses a parallel tree reduction in the log semiring.
  - Posteriors are taken as the gradient of log Z with respect to the potentials (forward–backward).

Inference

- `score_hmm_chromosome.py --score-source hmm` writes:
  - boundary-event marginals as site scores, indexed exactly like the raw heads;
  - with `--save-state-posteriors`, the per-base state posteriors `[L,7]`.
- For UniAnn, the state posteriors are collapsed to non-coding and frame k (section 2, last part).
- UniAnn (`uniann-pr45`) then decodes as before. **We do not use the CRF's own Viterbi** for gene models.

2. Model and loss in equations

Notation

- One window $x_{1:L}$, with $L = 10^4$. A path $y = (y_1, \dots, y_L)$ assigns each base a state $y_t \in \mathcal{S} = \{\text{IR}, E_0, E_1, E_2, I_0, I_1, I_2\}$.
- $\mathcal{Y}$ is the set of grammatical paths (the motif-gated transitions in section 1).
- $\mathcal{Y}_\ell \subseteq \mathcal{Y}$ is the subset compatible with the label ℓ:
  - single path: the longest CDS of each EviAnn gene;
  - arm C: the union of isoforms, with boundary events only at annotated positions;
  - $\ell_t = -1$ (unlabelled), and the 3 bases at each window edge, allow every state.
- Network outputs at position t (softmax probabilities):

Symbol	Head	Classes
$p_t^{\text{st}}(c)$	state head	$c \in \{\text{nc}, f_0, f_1, f_2, \text{intron}\}$
$p_t^{\text{sp}}(\cdot)$	splice head	{none, donor, acceptor}

Symbol	Head	Classes
$p_t^{\mathrm{ss}}(\cdot)$	start/stop head	{none, start, stop}
$p_t^{\mathrm{jf}}(\cdot)$	joint-frame head	$\{\mathrm{nc}, f_0, f_1, f_2\}$

CRF potentials (9 learnable scalars α, w_b, c_b)

Emission, where κ maps IR → nc, $E_k \to f_k$ and $I_k \to$ intron:

$$\psi_t(s) = \alpha \, \log p_t^{\mathrm{st}}(\kappa(s))$$

Boundary score, $b \in \{\mathrm{start}, \mathrm{stop}, \mathrm{donor}, \mathrm{acceptor}\}$:

$$\phi_t^b = w_b \log p_t^b + c_b$$

where $p_t^{\mathrm{start}} = p_t^{\mathrm{ss}}(\mathrm{start})$, $p_t^{\mathrm{donor}} = p_t^{\mathrm{sp}}(\mathrm{donor})$, and so on, indexed by motif start.

Transition into position t ($-\infty$ for every pair not listed):

$$T_t(s', s) = \begin{cases} \phi_t^{\mathrm{start}} & (\mathrm{IR}, E_0), \ \text{ATG at } t \\ \phi_{t-3}^{\mathrm{stop}} & (E_2, \mathrm{IR}), \ \text{stop codon at } t-3..t-1 \\ \phi_t^{\mathrm{donor}} & (E_k, I_{k+1 \bmod 3}), \ \text{GT at } t \ \text{(not after an in-frame stop)} \\ \phi_{t-2}^{\mathrm{acceptor}} & (I_k, E_k), \ \text{AG at } t-2..t-1 \\ 0 & (\mathrm{IR}, \mathrm{IR}), \ (I_k, I_k), \ (E_0, E_1), \ (E_1, E_2), \ (E_2, E_0) \ \text{unless a stop codon ends at } t-1 \end{cases}$$

Path score and partition functions:

$$S(y) = \sum_{t=1}^{L} \psi_t(y_t) + \sum_{t=2}^{L} T_t(y_{t-1}, y_t), \qquad Z = \sum_{y \in \mathcal{Y}} e^{S(y)}, \qquad Z_\ell = \sum_{y \in \mathcal{Y}_\ell} e^{S(y)}$$

Both sums are computed exactly by a tree reduction of $\prod_t M_t$, where $M_t(s', s) = T_t(s', s) + \psi_t(s)$, in the (logsumexp, +) semiring.

Loss

CRF term (per-base NLL of the label-constrained path set):

$$\mathcal{L}_{\mathrm{CRF}} = \frac{1}{L}(\log Z - \log Z_\ell) = -\frac{1}{L} \log P(y \in \mathcal{Y}_\ell \mid x)$$

Total loss, with $\lambda = 1$ for arms A and C and $\lambda = 0$ for B:

$$\mathcal{L} = 0.25 \, \mathcal{L}_{\mathrm{splice}} + \mathcal{L}_{\mathrm{ss}} + 0.1 \, \mathcal{L}_{\mathrm{jf}} + \lambda \, \mathcal{L}_{\mathrm{CRF}}$$

Per-position terms. $\mathcal{C}$ is the set of motif candidates (GT/AG, or ATG/stop) plus every labelled positive; $\mathcal{C}^{\mathrm{cds}}$ is the set of CDS bases with a known phase; $\mathcal{N}$ is the set of non-CDS bases.

$$\mathcal{L}_{\mathrm{splice}} = \frac{1}{|\mathcal{C}|} \sum_{t \in \mathcal{C}} -\log p_t^{\mathrm{sp}}(z_t), \qquad \mathcal{L}_{\mathrm{jf}} = \frac{1}{|\mathcal{C}^{\mathrm{cds}}|} \sum_{t \in \mathcal{C}^{\mathrm{cds}}} -\log p_t^{\mathrm{jf}}(f_{\phi_t}) + 0.1 \, \frac{1}{|\mathcal{N}|} \sum_{t \in \mathcal{N}} -\log p_t^{\mathrm{jf}}(\mathrm{nc})$$

$\mathcal{L}_{\mathrm{ss}}$ has the same form as $\mathcal{L}_{\mathrm{splice}}$, with the start/stop head.

Why this loss: its gradient

Write $\gamma_t(s) = P(y_t = s \mid x)$ for the unconstrained posterior and $\gamma_t^{\ell}(s) = P(y_t = s \mid x, y \in \mathcal{Y}_\ell)$ for the label-constrained one. Then

$$\frac{\partial \mathcal{L}_{\text{CRF}}}{\partial \psi_t(s)} = \frac{1}{L}\big(\gamma_t(s) - \gamma_t^{\ell}(s)\big), \qquad \frac{\partial \mathcal{L}_{\text{CRF}}}{\partial \phi_t^b} = \frac{1}{L}\big(\epsilon_t^b - \epsilon_t^{b,\ell}\big)$$

Here $\epsilon_t^b = P(\text{event } b \text{ at } t \mid x)$ is the boundary-event posterior, and $\epsilon_t^{b,\ell}$ is the same under the label constraint.

What this means:

- **Sequence-level normalisation.** The gradient is non-zero only where the model's *gene-parse* posterior disagrees with the label. A false GT that no competing parse uses gets no gradient. A false GT that produces a wrong gene model gets pushed down. Per-position cross-entropy treats every candidate independently.
- **Sum, not max.** Where $\ell_t = -1$, γ_t^{ℓ} is itself a posterior, so ambiguous bases are marginalised instead of forced. The training objective uses exactly the quantities we decode with: γ and ϵ.
- **Keep the per-position CE.** The CRF only sees motif-gated candidates, so the heads still need direct supervision. Keeping it also allows a raw-heads vs HMM-posterior comparison inside one model.
- **Few HMM parameters.** All position-specific evidence comes from the network, and the HMM only learns α, w_b, c_b. It cannot memorise a genome-specific transition matrix.
- **Longest-CDS labels by default.** One grammatical path per gene is unambiguous. The all-isoform $\mathcal{Y}_\ell$ (arm C) is the principled extension; so far it is about equal (74.2–74.4 vs 73.8–74.1 locus F1).

Inference: posteriors and the UniAnn inputs

Posteriors are obtained as gradients of $\log Z$ (equivalent to forward–backward):

$$\gamma_t(s) = \frac{\partial \log Z}{\partial \psi_t(s)}, \qquad \epsilon_t^b = \frac{\partial \log Z}{\partial \phi_t^b}$$

Inputs handed to UniAnn:

- site probability at a candidate t: ϵ_t^b (HMM arm) or p_t^b (raw arm);
- per-base frame: $p_t^{\text{nc}} = \gamma_t(\text{IR}) + \sum_k \gamma_t(I_k)$ and $p_t^{f_k} = \gamma_t(E_k)$; the raw arm uses p_t^{jf} instead.

UniAnn scores, from `prepare_sites.pl` and `pr_all.pl`:

$$\text{site}_t = \Big\lfloor \tfrac{1000}{\max_t \ln(e\, p_t^{\text{donor}})} + \tfrac{1}{2} \Big\rfloor \cdot \ln(e\, p_t^b + 10^{-10}) \;\approx\; 1000\,(1 + \ln p_t^b)$$

$$\text{nc}_t = \ln(e\, p_t^{\text{nc}} + 10^{-10}), \qquad \text{frame}_{k,t} = \ln(e\, p_t^{f_k}\,(1 - p_t^{\text{nc}}) + 10^{-10})$$

3. How this differs from other tools

Other tools' details below are from memory; please check them against the papers before citing.

	This work	Tiberius	Helixer + HelixerPost	AUGUSTUS	UniAnn (current pipeline)
NN	CNN-Mamba site + state heads	CNN + LSTM	CNN/BiLSTM per-base classes	none	none (consumes our scores)
HMM states	7 (IR, E0–2, I0–2), single strand	~15, explicit boundary states, both strands	HMM over base classes	generalized HMM with length distributions	own gene grammar
Transitions	position-dependent, from NN site heads, motif-gated	learned global matrix	fixed/heuristic	trained by counting	site scores from input files
HMM trained?	yes, end-to-end with the NN (9 scalars)	yes, end-to-end	no (post hoc)	yes, standalone	no

	This work	Tiberius	Helixer + HelixerPost	AUGUSTUS	UniAnn (current pipeline)
Training loss	CRF NLL of the label-constrained path set + per-position CE	loss on HMM-layer outputs vs labels	per-base CE only	ML counts	–
Decoding	posterior marginals → UniAnn Viterbi	its own HMM (Viterbi)	HelixerPost	Viterbi / sampling	Viterbi

Main differences from Tiberius:

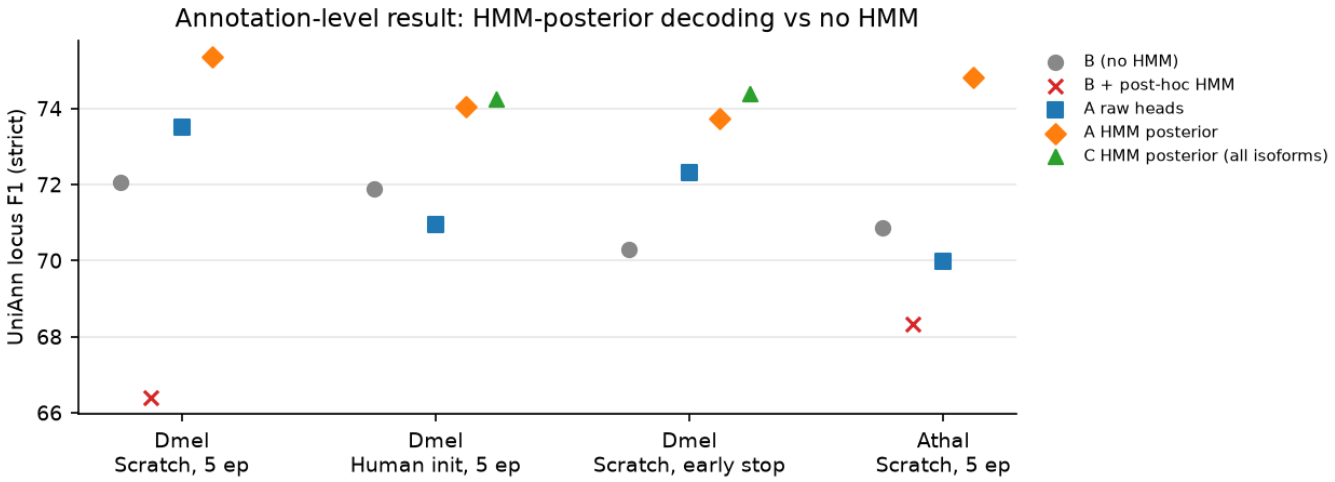
- Our transitions come **per position from the site heads**, instead of a global learned matrix.
- We have no explicit boundary states.
- We model a single strand; the minus strand is handled as reverse-complement windows.
- We hand the posterior to an **external decoder** (UniAnn) rather than decoding with the training HMM.

Against Helixer: our post-hoc control (row "B + post-hoc HMM" below) is the Helixer-style option, and it loses.

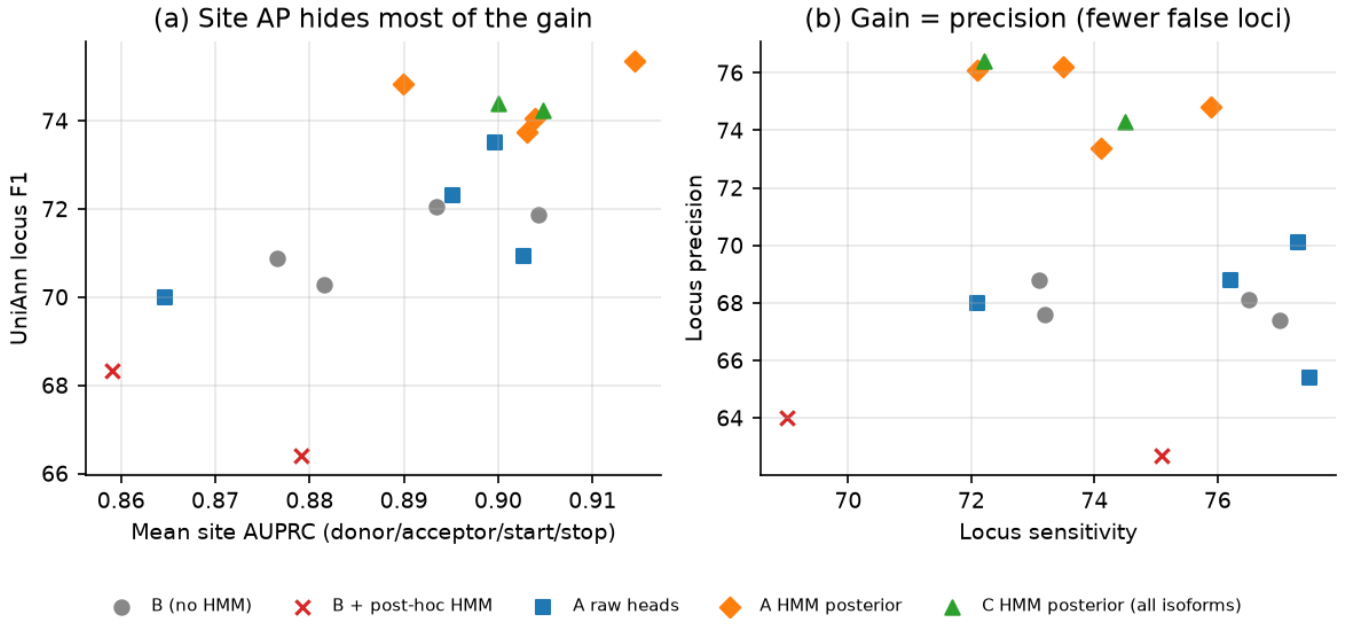
4. Results

Setup:

- Dmel: train/val on EviAnn pseudo-labels, val chr3L, test chrX plus strand against FlyBase.
- Athal: train chr1,2,4 / val chr3 / test chr5 plus strand against TAIR10.1.
- UniAnn evaluation follows our earlier validation of your pipeline, i.e. the `pr_all.pl` converter, `uniann-pr45`, and `gffcompare --strict-match -e 0`. Our reproduction of your pipeline matched exactly (locus F1 79.62).
- One seed per arm.



Species	Setting	Arm	Decode	Mean site AP	Locus Sn	Locus Pr	Locus F1	Transcript F1	Exon F1
Dmel	Scratch, 5 ep	B: no HMM	raw	0.8934	76.5	68.1	72.06	54.84	83.76
Dmel	Scratch, 5 ep	B + post-hoc HMM	hmm	0.8791	69.0	64.0	66.41	50.28	79.86
Dmel	Scratch, 5 ep	A: HMM loss	raw	0.8997	77.3	70.1	73.52	55.84	84.20
Dmel	Scratch, 5 ep	A: HMM loss	**hmm**	**0.9145**	75.9	74.8	**75.35**	**56.58**	**85.00**
Dmel	Human init, 5 ep	B: no HMM	raw	0.9043	77.0	67.4	71.88	54.82	84.15
Dmel	Human init, 5 ep	A: HMM loss	raw	0.9027	77.5	65.4	70.94	54.35	83.58
Dmel	Human init, 5 ep	A: HMM loss	hmm	0.9039	72.1	76.1	74.05	55.19	83.80
Dmel	Human init, 5 ep	C: all isoforms	hmm	0.9048	72.2	76.4	74.24	55.35	82.27
Dmel	Scratch, cosine + ES	B: no HMM	raw	0.8815	73.2	67.6	70.29	53.23	82.84
Dmel	Scratch, cosine + ES	A: HMM loss	raw	0.8951	76.2	68.8	72.31	54.92	83.74
Dmel	Scratch, cosine + ES	A: HMM loss	hmm	0.9030	74.1	73.4	73.75	55.41	84.67
Dmel	Scratch, cosine + ES	C: all isoforms	hmm	0.9000	74.5	74.3	74.40	55.82	84.65
Athal	Scratch, 5 ep	B: no HMM	raw	0.8766	73.1	68.8	70.88	58.18	85.94
Athal	Scratch, 5 ep	B + post-hoc HMM	hmm	0.8590	75.1	62.7	68.34	56.79	85.29
Athal	Scratch, 5 ep	A: HMM loss	raw	0.8647	72.1	68.0	69.99	57.49	85.22
Athal	Scratch, 5 ep	A: HMM loss	**hmm**	**0.8899**	73.5	76.2	**74.83**	**60.89**	85.77



Reading:

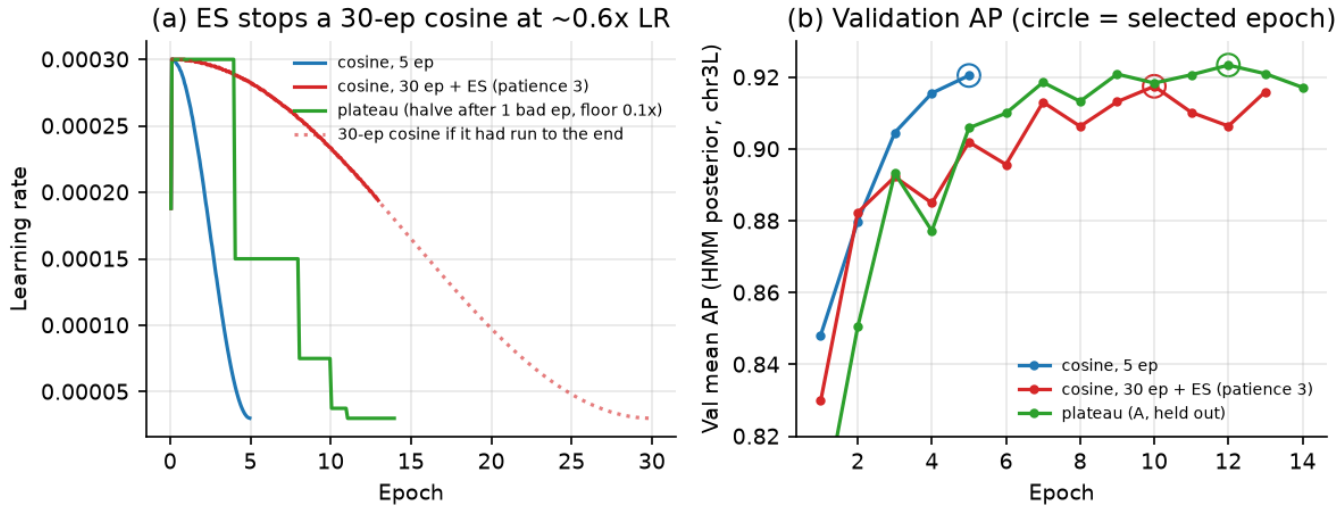
- **Arm A decoded through the HMM posterior beats B in all four settings.** Locus precision rises by 5–9 points. Sensitivity is flat or slightly down.
- **The gain is not in the raw heads.**
  - A's raw heads sit at or below B for Dmel human-init and for Athal.
  - Human initialisation lifts B's site AP by +0.011, but A-hmm still adds +2.2 locus F1 on top.
- **Post-hoc HMM hurts** (Dmel −5.7, Athal −2.5). In that control we train only the 9 HMM scalars and the state head on a frozen baseline for 2 epochs (`--train-only-hmm`). The heads must co-adapt with the CRF.
- **Site AUPRC is a poor proxy for annotation quality.** In panel (a), Athal A-hmm has a *lower* AP than several Dmel B points but the highest locus F1. Judge changes by UniAnn locus/transcript F1.

5. Training schedule: cosine + early stopping

- Schedule: linear warmup over $W = \max(20, \min(200, N/10))$ steps, where N is the total number of steps for the *requested* `--epochs` E. After warmup:

$$\eta(n) = \eta_0\big(0.1 + 0.45(1 + \cos \pi p_n)\big), \qquad p_n = \frac{n - W}{N - W}, \qquad \eta_0 = 3 \times 10^{-4}$$

- The decay is fixed in advance by `--epochs`. Setting a large cap (30) and relying on early stopping (patience 3) therefore means training stops while the LR is still about 0.6 $\eta 0$. The model never anneals.
  - Those runs stopped at epochs 13–14.
  - They were **worse** than plain 5-epoch runs: A-hmm locus F1 73.75 vs 75.35; site AP 0.903 vs 0.915.



Fix: added `--lr-schedule plateau`, a ReduceLROnPlateau-type schedule.

- Rule, with LR scale σ (initially 1) and b the count of consecutive non-improving epochs:

$$\eta = \eta_0\,\sigma, \qquad \text{if } b \geq 1 \text{ and } \sigma > 0.1: \quad \sigma \leftarrow \max(0.1,\ 0.5\,\sigma),\ \ b \leftarrow 0$$

- An epoch improves if validation selection AP exceeds the best so far by at least 10^{-4}.
- Training stops when $\sigma = 0.1$ and $b \geq 2$.
- Keep the best validation checkpoint. Checkpoint and resume restore the schedule state.

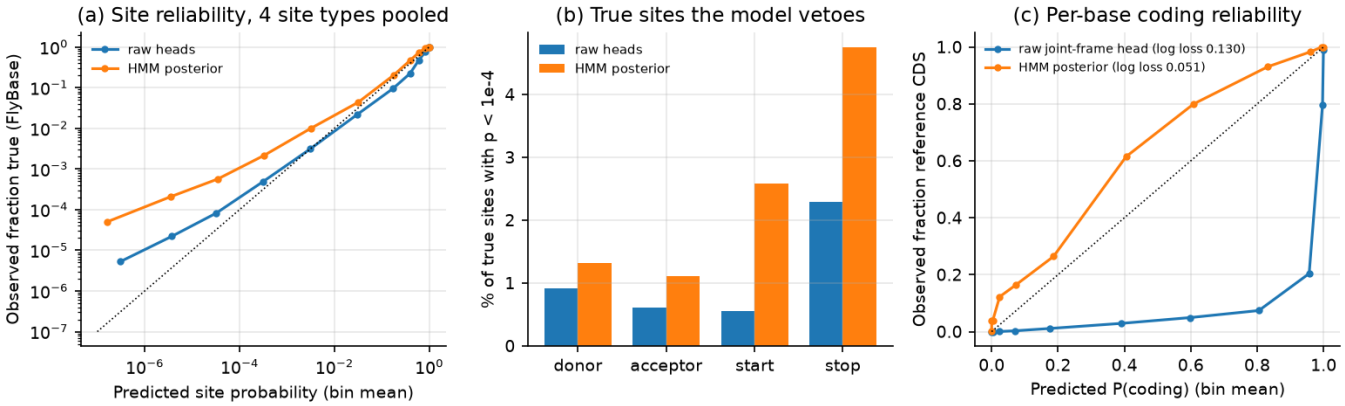
Dmel held-out runs with this schedule:

- A stopped after 14 epochs, with the best checkpoint at epoch 12.
  - Test mean AP: 0.9202 (HMM posterior) and 0.9131 (raw heads), vs 0.9145 for the 5-epoch cosine.
  - UniAnn locus F1: 75.4 (HMM posterior, the same as the 5-epoch cosine) and 74.3 (raw heads).
- B is still running.

A full-length 30-epoch cosine without early stopping (`chrxtrain`, section 0) anneals properly and gives the best result (80.6), selecting epoch 18 by validation AP.

Recommendation: use the plateau schedule, or a fixed full-length cosine. Do not combine a long cosine with patience-based early stopping.

6. Caveats and known issues



All of the following are measured on Dmel chrX+, scratch 5-ep model A, against FlyBase.

1. **Low-tail overconfidence of the HMM posterior (panels a, b).**
   - The posterior is roughly calibrated, or even slightly conservative, at the high end: candidates scored 0.5–0.9 are true 83–91% of the time.

- In the low tail it is too sure that a candidate is *not* a site. Candidates scored 1e-6–1e-4 are true at 1.6e-4 to 9.6e-4.
- True sites with p < 1e-4, raw → HMM:

Site	Raw	HMM
donor	0.9%	1.3%
acceptor	0.6%	1.1%
start	0.6%	2.6%
stop	2.3%	4.7%

- `prepare_sites.pl` scores sites as about $1000(1 + \ln p)$ (section 2), so these sites get scores around −10,000. That is effectively a hard veto in UniAnn, which fits the small sensitivity loss.
- Cheap fixes to try:
  - a probability floor: $\tilde{p} = \max(p, \varepsilon)$;
  - a mix with the raw heads: $\tilde{p} = (1 - \beta)\, \epsilon_t^b + \beta\, p_t^b$, e.g. $\beta = 0.1$;
  - a posterior temperature $\tau > 1$ on $S(y)/\tau$.

2. **Per-base coding (panel c).**
   - The **raw joint-frame head is badly overconfident toward coding**: bases scored 0.9–0.99 are reference CDS only 20% of the time. A guess is that it fires on minus-strand CDS or UTR; this is unverified.
   - Log loss is $\mathrm{LL} = -\frac{1}{L} \sum_t [c_t \log q_t + (1 - c_t) \log(1 - q_t)]$, where $q_t = P(\text{coding})$ and c_t indicates a reference CDS base.
   - The HMM posterior is much better (log loss 0.051 vs 0.130). However, it still puts $P(\text{coding}) < 0.01$ on 3.8% of reference CDS bases, i.e. it misses whole genes.
3. **`pr_all.pl` double counting.**
   - The frame columns we pass are already joint probabilities: $p_t^{\text{nc}} + \sum_k p_t^{f_k} = 1$.
   - The converter's `frame` formula (section 2) therefore uses $P(\text{coding})^2$:

$$p_t^{f_k}\,(1 - p_t^{\text{nc}}) = P(f_k)\,P(\text{coding}), \qquad \text{whereas the intended value is } \ln(e\, p_t^{f_k})$$

- Example: $p^{\text{nc}} = p^{f_k} = 0.5$ gives $\ln(0.25\, e)$ instead of $\ln(0.5\, e)$.
- The fix would be to pass the conditional $p_t^{f_k} / (1 - p_t^{\text{nc}})$ instead.
- All arms, and your original pipeline, share this, so the comparisons above are fair. Please check which convention UniAnn expects.

4. **Evaluation scope.** Plus strand of one test chromosome per species, one seed per arm.
5. **Human-init fine-tune.** At 3e-4, validation HMM AP is unstable in epochs 1–2. The fresh state head disagrees with the confident pretrained emissions. A lower fine-tune LR is untested.

7. Suggested next steps

1. Re-run UniAnn with (a) a posterior floor or a raw mix for sites, and (b) conditional frame probabilities in `pr_all`. Each Dmel/Athal UniAnn run takes about 2 min.
2. Seeds: repeat the 30-epoch `chrxtrain` gene-HMM run and your b010 recipe with 2–3 seeds, to tell whether 80.6 vs 80.0 is a real difference.
3. Finish the plateau-schedule B arm (held-out).
4. Score the minus strand and the remaining chromosomes. Add another species (EviAnn GFFs exist for Mmus, Ggal, Drer and Ptri; they need HPC).
5. Optionally, decode with the CRF's own Viterbi (`GeneHMM.decode`) as a UniAnn-free comparison.
6. **Calibration: please push your vector-scaling code.** The only calibration on `origin/main` is scalar temperature scaling (809f743, "Publish Human four-state model and calibration"); we could not find a vector-scaling commit. Once it is pushed, it can be applied to the HMM-posterior sites as well. This is relevant to the low-tail overconfidence in section 6.

8. Reproduction

- Code: branch `gene-hmm-isoforms` (see the top of this note). Main additions over `main`:
  - `src/cnn_mamba/gene_hmm.py`: the CRF layer (partition function, posteriors, Viterbi);

  - `train.py`: `--hmm-loss-weight`, `--selection-metric hmm`, `--lr-schedule plateau`, `--seed`, `--train-only-hmm`, `--init-checkpoint`;
  - `score_hmm_chromosome.py`: `--score-source hmm|raw`, `--save-state-posteriors`;
  - `prepare_droso.py`: `--include-path-labels`, `--path-isoforms all`, and `--chromosomes` with a JSON chromosome map (Athal: `configs/atha_chromosomes.json`);
  - `evaluate_candidates.py`: non-Dmel chromosomes.
- On a 12 GB GPU use `--batch-size 2 --grad-accum 4 --gradient-checkpointing`.
- Common training args:

  ```
  --mode baseline --architecture cnn_mamba --cnn-kernel-size 7 --window-size 10000 --stride 5000
  --frame-dilation 3 --joint-frame-aux-weight 0.1 --joint-noncoding-weight 0.1 --lr 3e-4
  --reference-name FlyBase --test-name chrX
  ```

  - Arm A adds `--hmm-loss-weight 1.0 --selection-metric hmm`; arm B is the same without it.
  - Arm C additionally uses data prepared with `--path-isoforms all`.
  - Data preparation needs `--include-path-labels`.
  - `chrxtrain` adds `--include-test-in-train`. The 80.6 run uses `--epochs 30 --early-stopping-patience 0` (cosine, run to the end, best validation checkpoint kept).
  - Plateau schedule: `--lr-schedule plateau --plateau-patience 1 --plateau-factor 0.5 --min-lr-ratio 0.1 --early-stopping-patience 2 --early-stopping-min-delta 0.0001`.
- Scoring chrX: `score_hmm_chromosome.py --score-source hmm --save-state-posteriors` for the HMM arm, `score_joint_frame_chromosome.py` plus `score_hmm_chromosome.py --score-source raw` for raw heads.
- UniAnn: site files from `prepare_sites.pl`, frame file from `pr_all.pl` (state posteriors collapsed as in section 2), then `uniann-pr45` and `gffcompare --strict-match -e 0` against the plus-strand reference.
- Figures: `make_figures.py` in this folder; the input tables are in `data/` (`compare.csv` holds the full metric table). Figures 3–4 also need the training logs and score arrays (see the script's docstring).